

ELK ISLAND NATL PARK, AB, Canada

WMO: 712380

Lat: 53.6825N

Lon: 112.8681W

Elev: 716

StdP: 93.01

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.3	-28.8	-36.2	0.1	-32.2	-32.6	0.2	-28.6	6.7	-10.7	6.0	-11.9	0.5	130	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	28.2	18.2	26.5	17.2	24.9	16.3	19.8	25.8	18.5	24.6	17.4	23.1	1.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	13.7	23.4	16.2	12.5	21.5	15.1	11.7	20.1	59.9	25.8	55.4	24.6	51.7	23.2	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.3	5.3	4.5	-37.4	31.5	4.1	2.2	-40.3	33.0	-42.7	34.3	-45.1	35.5	-48.1	37.1		
			-37.4	22.1	4.0	1.5	-40.3	23.2	-42.6	24.1	-44.9	24.9	-47.8	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-11.4	-9.6	-4.4	3.8	10.7	14.6	17.0	15.9	11.0	4.3	-4.1	-10.3
	DBStd	11.95	9.05	8.03	7.83	5.28	4.58	3.20	3.03	3.47	4.46	4.73	6.75	8.14
	HDD10.0	3184	662	550	446	193	47	3	1	2	41	184	424	631
	HDD18.3	5573	921	783	704	436	239	118	62	89	222	436	674	889
	CDD10.0	697	0	0	0	7	68	141	218	186	70	7	0	0
	CDD18.3	47	0	0	0	0	2	6	22	15	2	0	0	0
	CDH23.3	708	0	0	0	1	52	99	283	208	63	2	0	0
	CDH26.7	138	0	0	0	0	7	14	64	41	12	0	0	0
Wind	WSAvg	1.7	1.5	1.6	2.0	2.3	2.1	1.7	1.5	1.4	1.5	1.9	1.7	1.5
Precipitation	PrecAvg	465	24	13	22	34	51	79	94	51	38	22	19	20
	PrecMax	645	68	27	55	68	125	153	187	111	82	57	44	50
	PrecMin	274	4	2	6	1	8	26	43	14	12	3	2	6
	PrecStd	77	17	8	12	20	26	36	32	24	21	15	13	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	10.0	13.6	21.8	27.9	28.4	31.1	30.2	28.4	21.8	12.6	8.9
		MCWB	3.5	3.3	6.2	10.1	14.1	16.8	19.8	20.1	16.8	12.4	6.4	3.1
	2%	DB	5.7	6.7	10.5	18.5	24.7	26.1	28.3	27.6	25.0	17.1	8.6	5.5
		MCWB	1.9	2.0	4.3	8.7	13.2	15.9	19.1	18.6	15.1	9.6	4.1	1.7
	5%	DB	3.6	4.1	8.2	15.3	22.2	24.1	26.4	25.7	22.3	14.2	6.2	3.3
		MCWB	0.5	0.6	3.1	7.0	12.0	15.1	18.1	17.7	14.2	7.9	2.5	0.1
	10%	DB	1.1	1.6	5.8	12.7	19.7	22.1	24.6	23.7	19.2	11.7	4.0	0.8
		MCWB	-1.0	-0.9	1.8	5.7	10.7	14.2	17.2	16.7	12.5	6.6	1.2	-1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	4.2	6.5	10.7	16.4	18.8	21.9	21.9	17.9	12.5	6.7	3.7
		MCDB	7.9	9.1	12.7	20.3	23.2	25.2	28.2	28.4	25.6	20.4	12.2	8.0
	2%	WB	2.2	2.3	4.7	9.1	14.3	17.4	20.2	19.8	16.1	10.3	4.5	1.8
		MCDB	5.5	6.1	9.6	17.5	22.3	23.6	26.3	25.5	23.3	15.6	7.9	5.3
	5%	WB	0.7	0.7	3.4	7.6	12.8	16.2	18.8	18.4	14.6	8.6	2.9	0.3
		MCDB	3.4	3.9	7.7	14.3	20.5	22.1	25.0	24.3	21.2	13.1	5.7	3.0
	10%	WB	-1.1	-0.9	2.0	6.2	11.5	15.1	17.7	17.1	13.1	7.1	1.4	-1.4
		MCDB	1.1	1.8	5.6	11.7	18.5	20.9	23.4	22.7	18.6	11.0	3.7	0.9
Mean Daily Temperature Range	5% DB	MDBR	11.3	13.1	12.6	12.2	12.7	12.0	12.3	12.6	12.5	10.6	9.4	10.5
		MCDBR	12.9	16.3	15.0	16.2	16.4	15.3	15.3	15.7	17.6	15.3	11.7	11.9
		MCWBR	9.5	12.0	9.6	8.4	7.5	7.2	7.5	7.9	8.8	8.3	7.6	8.7
	5% WB	MCDBR	12.6	15.7	14.2	15.2	14.9	13.2	14.0	14.4	16.4	14.3	10.9	11.4
		MCWBR	9.4	11.6	9.2	8.3	7.3	7.0	7.5	7.8	8.5	8.2	7.4	8.7
Clear-Sky Solar Irradiance	taub	0.254	0.258	0.309	0.326	0.339	0.353	0.353	0.361	0.305	0.277	0.251	0.242	
	taud	2.307	2.365	2.304	2.383	2.396	2.375	2.416	2.349	2.501	2.551	2.449	2.281	
	Ebn at Noon	734	864	875	907	908	895	889	858	878	838	750	671	
	Edh at Noon	56	74	99	105	110	114	107	108	81	62	49	47	
All-Sky Solar Radiation	RadAvg	0.96	2.00	3.54	4.67	5.72	6.01	6.14	5.05	3.49	2.00	0.95	0.70	
	RadStd	0.11	0.15	0.27	0.43	0.43	0.39	0.30	0.35	0.36	0.24	0.11	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page